

## Fall 2025, 4100 Problem Set 3, Hayden Fuller

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Given that  $\mathbf{b}_1$  and  $\mathbf{b}_2$  are two nonzero, linear independent vectors in  $\mathbb{R}^3$  and that  $T \in \mathcal{L}(\mathbb{R}^4, \mathbb{R}^3)$  is such that

$$T\mathbf{e}_1 = \mathbf{b}_1, \quad T\mathbf{e}_2 = -\mathbf{b}_1, \quad T\mathbf{e}_3 = 2\mathbf{b}_1, \quad \text{and} \quad T\mathbf{e}_4 = \mathbf{b}_2$$

find the general solution to  $T\mathbf{x} = \mathbf{b}_1$

$$\mathbf{x} = (x_1, x_2, x_3, x_4)$$

$$T\mathbf{x} = x_1\mathbf{b}_1 + x_2(-\mathbf{b}_1) + x_3(2\mathbf{b}_1) + x_4\mathbf{b}_2 = (x_1 - x_2 + 2x_3)\mathbf{b}_1 + x_4\mathbf{b}_2$$

since  $\mathbf{b}_1, \mathbf{b}_2$  are linearly independent,  $T\mathbf{x} = \mathbf{b}_1$  iff  $x_1 - x_2 + 2x_3 = 1, x_4 = 0$

parameterize free variables  $s = x_2, t = x_3$

$$x_1 = 1 + s - 2t, \quad x_4 = 0$$

$$\mathbf{x} = (1 + s - 2t, s, t, 0), \quad s, t \in \mathbb{R}$$

$$\mathbf{x} = (1, 0, 0, 0) + s(1, 1, 0, 0) + t(-2, 0, 1, 0), \quad s, t \in \mathbb{R}$$

an affine 2-plane in  $\mathbb{R}^4$

2. Let  $V$  be the vector space  $\mathbb{P}_3(\mathbb{R})$  endowed with the inner product,  $\langle x^i, x^j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$  and let  $S$  be a subspace of  $V$  given by,  $S = \{f \in \mathbb{P}_3 : f(1) = f'(1) = 0\}$ . Find bases for  $S$  and  $S^\perp$ .

$$V = \mathbb{P}_3(\mathbb{R})$$

$$\text{monomial basis } \{1, x, x^2, x^3\}$$

$$f(x) = a + bx + cx^2 + dx^3$$

inner product makes monomial orthonormal so

$$f, g \text{ with vectors } (a, b, c, d) \text{ and } (a\tilde{a} + b\tilde{b} + c\tilde{c} + d\tilde{d})$$

$$\langle f, g \rangle = a\tilde{a} + b\tilde{b} + c\tilde{c} + d\tilde{d}$$

$$S = \{f \in \mathbb{P}_3 : f(1) = f'(1) = 0\} \text{ constraints:}$$

$$f(1) = a + b + c + d = 0, \quad f'(1) = b + 2c + 3d = 0.$$

From  $b + 2c + 3d = 0$  we get

$$b = -2c - 3d$$

$$a = -(b + c + d) = c + 2d$$

$$\text{coefficient vector: } (a, b, c, d) = (c + 2d, -2c - 3d, c, d) = c(1, -2, 1, 0) + d(2, -3, 0, 1).$$

$$\dim S = 2$$

$$\text{a convenient basis of } S \text{ is } S = \text{span}\{1 - 2x + x^2, 2 - 3x + x^3\}$$

$S^\perp$  is the orthogonal complement in  $\mathbb{R}^4$

Let a general polynomial in  $S^\perp$  have coefficients  $(\alpha, \beta, \gamma, \delta)$

$$\alpha - 2\beta + \gamma = 0, \quad 2\alpha - 3\beta + \delta = 0$$

$$\beta = 2\gamma - \delta, \quad \alpha = 3\gamma - 2\delta$$

$$(\alpha, \beta, \gamma, \delta) = \gamma(3, 2, 1, 0) + \delta(-2, -1, 0, 1)$$

$$\dim S^\perp = 2$$

$$\text{basis is } S^\perp = \text{span}\{3 + 2x + x^2, -2 - x + x^3\}$$

**Bonus:** Find an integral formula for the above inner product

3. Let  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{m \times l}$ . Show that  $\text{range}(A)$  is orthogonal to  $\text{range}(B)$  if and only if  $A^T B$  is the zero matrix.

Let the columns of  $A$  be  $a_1, \dots, a_n \in \mathbb{R}^m$  and the columns of  $B$  be  $b_1, \dots, b_l \in \mathbb{R}^m$

$$\text{range}(A) = \text{span}\{a_1, \dots, a_n\}$$

$$\text{range}(B) = \text{span}\{b_1, \dots, b_l\}$$

$$(A^T B)_{ij} = a_i^T b_j$$

If  $\text{range}(A)$  is orthogonal to  $\text{range}(B)$ , then every column  $a_i$  is orthogonal to every column  $b_j$

$$a_i^T b_j = 0 \text{ for all } i, j$$

every  $A^T B$  is zero, so  $A^T B = 0$

if  $A^T B = 0$  then  $a_i^T b_j = 0$  for all  $i, j$

any  $x \in \text{range}(A)$  and  $y \in \text{range}(B)$  are linear combinations  $x = \sum_i \alpha_i a_i$ ,  $y = \sum_j \beta_j b_j$

their inner product is  $x^T y = \sum_{i,j} \alpha_i \beta_j a_i^T b_j = 0$

every vector in  $\text{range}(A)$  is orthogonal to every vector in  $\text{range}(B)$

$\text{range}(A) \perp \text{range}(B)$

4. Compute the fundamental subspaces of the following matrix

$$A = \begin{bmatrix} 1 & 0 & 4 & -3 & 2 \\ 4 & -2 & 1 & -12 & 3 \\ -3 & 2 & 3 & 9 & -1 \end{bmatrix}$$

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 & 4 & -3 & 2 \\ 0 & 1 & 15/2 & 0 & 5/2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A) = 2$$

1) Column space  $\text{range}(A) \subset \mathbb{R}^3$

Pivot columns are 1 and 2

basis given by first two columns of  $A$

$$\text{range}(A) = \text{span}\{ [1, 4, -3]^T, [0, -2, 2]^T \}$$

$$\dim \text{range}(A) = 2$$

2) Row space  $\text{row}(A) \subset \mathbb{R}^5$

basis is two independent rows of  $A$

$$\text{row}(A) = \text{span}\{ [1, 0, 4, -3, 2], [4, -2, 1, -12, 3] \}$$

$$\dim \text{row}(A) = 2$$

3) Nullspace  $\mathcal{N}(A) = \{x \in \mathbb{R}^5 : Ax = 0\}$

From RREF free variables are  $x_3, x_4, x_5$

$$\text{basis } \mathcal{N}(A) = \text{span} \left\{ \begin{bmatrix} -4 \\ -15/2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -5/2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\dim \mathcal{N}(A) = 5 - \text{rank}(A) = 3$$

4) Left nullspace  $\mathcal{N}(A^T) = \{y \in \mathbb{R}^3 : A^T y = 0\}$

Find  $A^T$  nullspace

$$\mathcal{N}(A^T) = \text{span}\{ [-1, 1, 1]^T \}$$

$$\dim \mathcal{N}(A^T) = 3 - \text{rank}(A) = 1$$

5. Let

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

a. Draw a directed graph having  $A$  as its incidence matrix

$e_1 : 2 \rightarrow 1$  (column 1 has  $+1$  in row 1 and  $-1$  in row 2)

$e_2 : 1 \rightarrow 3$  (column 2:  $-1$  at row 1,  $+1$  at row 3)

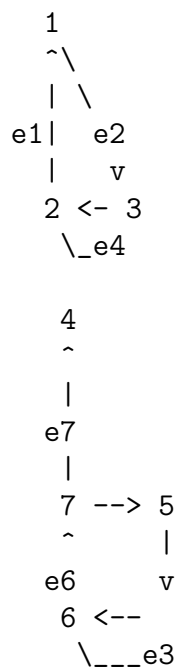
$e_3 : 5 \rightarrow 6$

$e_4 : 3 \rightarrow 2$

$e_5 : 7 \rightarrow 5$

$e_6 : 6 \rightarrow 7$

$e_7 : 7 \rightarrow 4$



- b. Find bases for the four fundamental subspaces;  $\text{range}(A)$ ,  $\text{range}(A^T)$ ,  $\text{null}(A)$ , and  $\text{null}(A^T)$

$m$  vertices and  $c$  connected components the rank is  $m - c$

$$m = 7, c = 2, \text{rank}(A) = 7 - 2 = 5$$

$$\dim \text{range}(A) = 5$$

$$\dim \text{null}(A) = n - \text{rank} = 7 - 5 = 2$$

$$\dim \text{null}(A^T) = m - \text{rank} = 2$$

1)  $\text{null}(A)$  (edge-cycle space)

There are two independent cycles here:

1-3-2-1 uses edges  $e_1, e_2, e_4$

Put 1 on those three edge-positions and 0 elsewhere:

$$v_1 = [1, 1, 0, 1, 0, 0, 0]^T$$

cycle on vertices 5-6-7-5 uses edges  $e_3, e_6, e_5$ :

$$v_2 = [0, 0, 1, 0, 1, 1, 0]^T$$

these are linearly independent and  $\text{span } \text{null}(A) = \text{span}\{v_1, v_2\}$

2)  $\text{null}(A^T)$  (vertex-potentials constant on components)

$A^T y = 0$  means for every edge  $j$ ,  $y_{\text{head}(j)} - y_{\text{tail}(j)} = 0$

$y$  is constant on each connected component

indicator of component  $\{1, 2, 3\}$  is  $u_1 = [1, 1, 1, 0, 0, 0, 0]^T$

indicator of component  $\{4, 5, 6, 7\}$  is  $u_2 = [0, 0, 0, 1, 1, 1, 1]^T$

$$\text{null}(A^T) = \text{span}\{u_1, u_2\}$$

3)  $\text{range}(A)$  (column space)

$\text{range}(A)$  is the orthogonal complement of  $\text{null}(A^T)$

consists of all vectors in  $\mathbb{R}^7$  whose entries sum to zero on each connected component

basis for component  $\{1, 2, 3\}$ :  $r_1 = [1, -1, 0, 0, 0, 0, 0]^T$ ,  $r_2 = [0, 1, -1, 0, 0, 0, 0]^T$

basis for component  $\{4, 5, 6, 7\}$ :  $r_3 = [0, 0, 0, 1, -1, 0, 0]^T$ ,  $r_4 = [0, 0, 0, 0, 1, -1, 0]^T$ ,  $r_5 = [0, 0, 0, 0, 0, 1, -1]^T$

These 5 vectors are independent and span the column space  $\text{range}(A) = \text{span}\{r_1, r_2, r_3, r_4, r_5\}$

4)  $\text{range}(A^T)$  (row space of  $A$ )

$\text{range}(A^T)$  is the orthogonal complement of  $\text{null}(A)$

dimension 5

basis is given by any 5 independent rows of  $A$

$$\text{row}_1 = [1, -1, 0, 0, 0, 0, 0]$$

$$\text{row}_2 = [-1, 0, 0, 1, 0, 0, 0]$$

$$\text{row}_3 = [0, 1, 0, -1, 0, 0, 0]$$

$$\text{row}_5 = [0, 0, -1, 0, 1, 0, 0]$$

$$\text{row}_6 = [0, 0, 1, 0, 0, -1, 0]$$

$$\text{range}(A^T) = \text{span}\{\text{row}_1, \dots, \text{row}_6 \setminus \{\text{row}_4\}\}$$

6. Let  $T$  be the linear map on  $\mathbb{R}^2$  defined as follows

$$\begin{aligned} T \begin{bmatrix} 1 \\ 0 \end{bmatrix} &= \begin{bmatrix} -2 \\ 1 \end{bmatrix} \\ T \begin{bmatrix} 2 \\ 1 \end{bmatrix} &= \begin{bmatrix} 0 \\ 4 \end{bmatrix} \end{aligned}$$

a. Find  $[T]_{\{e_i\}}$  the matrix representation of  $T$  with respect to the standard basis.

Let  $A = [T]_{\{e_i\}}$ . We are given

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

the first column of  $A$  is  $[-2, 1]^T$

columns of  $A$  as  $c_1, c_2$

second equation says  $2c_1 + c_2 = [0, 4]^T$

Using  $c_1 = [-2, 1]^T$

$$2[-2, 1]^T + c_2 = [0, 4]^T \implies c_2 = [0, 4]^T - [-4, 2]^T = [4, 2]^T$$

$$[T]_{\{e_i\}} = A = \begin{bmatrix} -2 & 4 \\ 1 & 2 \end{bmatrix}$$

b. Let  $\{\mathbf{v}_1, \mathbf{v}_2\}$  be another basis for  $\mathbb{R}^2$  where  $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$  and  $\mathbf{v}_2 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ . Find  $[T]_{\{\mathbf{v}_i\}}$ , the matrix representation of  $T$  with respect to the basis  $\{\mathbf{v}_i\}$  and use it write  $T(4\mathbf{v}_1 + 5\mathbf{v}_2)$  as a linear combination of the basis vectors,  $\mathbf{v}_i$ .

$$\mathbf{v}_1 = [1, -1]^T, \quad \mathbf{v}_2 = [2, -3]^T$$

$$\text{Form } P = [\mathbf{v}_1 \ \mathbf{v}_2] = \begin{bmatrix} 1 & 2 \\ -1 & -3 \end{bmatrix}$$

$$[T]_{\{\mathbf{v}_i\}} = P^{-1}AP$$

$$\text{Compute } P^{-1}. \det P = 1 \cdot (-3) - 2 \cdot (-1) = -1$$

$$P^{-1} = \frac{1}{-1} \begin{bmatrix} -3 & -2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ -1 & -1 \end{bmatrix}$$

$$AP = \begin{bmatrix} -2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & -3 \end{bmatrix} = \begin{bmatrix} -6 & -16 \\ -1 & -4 \end{bmatrix}$$

$$[T]_{\{\mathbf{v}_i\}} = P^{-1}(AP) = \begin{bmatrix} 3 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -6 & -16 \\ -1 & -4 \end{bmatrix} = \begin{bmatrix} -20 & -56 \\ 7 & 20 \end{bmatrix}$$

$$[T]_{\{\mathbf{v}_i\}} = \begin{bmatrix} -20 & -56 \\ 7 & 20 \end{bmatrix}$$

Apply  $T$  to  $4\mathbf{v}_1 + 5\mathbf{v}_2$

express in the  $\{\mathbf{v}_i\}$ -basis

coordinate vector of  $w = 4\mathbf{v}_1 + 5\mathbf{v}_2$  relative to  $\{\mathbf{v}_i\}$  is  $[4 \ 5]^T$

Then the coordinate vector of  $T(w)$  in that basis is

$$\begin{aligned}
[T]_{\{\mathbf{v}_i\}} \begin{bmatrix} 4 \\ 5 \end{bmatrix} &= \begin{bmatrix} -20 & -56 \\ 7 & 20 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} -20 \cdot 4 - 56 \cdot 5 \\ 7 \cdot 4 + 20 \cdot 5 \end{bmatrix} = \begin{bmatrix} -80 - 280 \\ 28 + 100 \end{bmatrix} = \begin{bmatrix} -360 \\ 128 \end{bmatrix} \\
T(4\mathbf{v}_1 + 5\mathbf{v}_2) &= -360 \mathbf{v}_1 + 128 \mathbf{v}_2
\end{aligned}$$

7. Let  $T : \mathbb{P}_2 \rightarrow \mathbb{R}_3$  be defined by  $T(f) = f(-2)\mathbf{e}_1 + f(1)\mathbf{e}_2 + f(3)\mathbf{e}_3$ . Find the matrix representation of  $T$  with respect to the standard bases of each space and use it to find the equation of the parabola passing through the points,  $(-2, 8), (1, 3), (3, -5)$ .

The standard basis of  $\mathbb{P}_2$  is  $\{1, x, x^2\}$

$$f(-2) = a - 2b + 4c, \quad f(1) = a + b + c, \quad f(3) = a + 3b + 9c$$

$$\text{since } [f(-2), f(1), f(3)]^T = [T][a \ b \ c]^T$$

$$\text{find the parabola through } (-2, 8), (1, 3), (3, -5) \text{ solve } \begin{bmatrix} 1 & -2 & 4 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \\ -5 \end{bmatrix}$$

$$a = \frac{28}{5}, \quad b = -\frac{32}{15}, \quad c = -\frac{7}{15}$$

$$f(x) = \frac{28}{5} - \frac{32}{15}x - \frac{7}{15}x^2 = \frac{1}{15}(84 - 32x - 7x^2)$$